SEAT NO. CT-21090

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(Bachelor of Science in Computer Science & Information Technology)

FALL SEMESTER EXAMINATIONS 2023

Batch 2021

Time : 3 Hours

Miss Yumna Sajid

Dated : 24-JAN-24

Max Marks : 60

Probability & Statistics - MT-331

INSTRUCTIONS:

- *Attempt **ALL** questions.*
- *Statistical Annexure is attached on last page.*
- *Use of scientific calculator is allowed.*

Q.1 (a) To evaluate effectiveness of a processor for a certain type of tasks, we recorded the CPU time for n = 30 randomly chosen jobs (in seconds), **(8)** **CLO-1**

70 36 43 69 82 48 34 62 35 15 59 139 46 37 42 30 55 56 36 82 38 89 54 25 35 24 22 9 56 19

i. What information do we get from this collection of numbers?

ii. Calculate Skewness and comment on whether or not there is any clear indication that the sample came from a population that has a bell-shaped distribution.

(b) Out of six computer chips, two are defective. If two chips are randomly chosen for testing (without replacement), compute the probability that both of them are defective. List all the outcomes in the sample space. **(4)** **CLO-1**

Q.2 (a) Let X be a continuous random variable with probability density function given by **(4)** **CLO-2**

$$f(x) = \begin{cases} k(x+1) & for\ 0 \le x \le 1 \\ 0 & ew \end{cases}$$

i. Find the value of k.

ii. Find E(X) and V(X).

(b) As part of a business strategy, randomly selected 20% of new internet service subscribers receive a special promotion from the provider. A group of 10 neighbors signs for the service. **(4)** **CLO-2**

i. What is the probability that at least 4 of them get a special promotion?

ii. What is the probability that at most 2 of them get a special promotion?

iii. What is the probability that none of them get a special promotion?

(c) The speed of a file transfer from a server on campus to a personal computer at a student's home on a weekday evening is normally distributed with a mean of 60 kilobits per second and a standard deviation of 4 kilobits per second. **(4)** **CLO-2**

i. What is the probability that the file will transfer at a speed of 70 kilobits per second or more?

ii. What is the probability that the file will transfer at a speed of less than 58 kilobits per second?

Q.3 (a) When a computer disk manufacturer tests a disk, it writes to the disk and then tests it using a certifier. The certifier counts the number of missing pulses or errors. The number of errors on a test area on a disk has a Poisson distribution with average 0.2. **(4)** **CLO-2**

i. What is the probability of test areas have at least one errors?

ii. What is the probability of test areas have two or fewer errors?

(b) List all possible sample of size 2 that can be selected from the population 2, 4, 6, 8 & 10 by using sampling without replacement. Calculate sample mean for each sample and then verify the following results: $E(\bar{x}) = \mu$ **(8)** **CLO-2**

$$v(\bar{x}) = \frac{N-n}{N-1}\left(\frac{\sigma^2}{n}\right)$$

Q.4 **(a)** Installation of a certain hardware takes random time with a standard deviation of 5 minutes. A computer technician installs this hardware on 64 different computers, with the average installation time of 42 minutes. Compute a 95% confidence interval for the population mean installation time. **(4)** **CLO-2**

(b) The McFarland Insurance Company Claims Department reports the mean cost to process a claim is $60. An industry comparison showed this amount to be larger than most other insurance companies, so the company instituted cost-cutting measures. To evaluate the effect of the cost-cutting measures, the Supervisor of the Claims Department selected a random sample of 26 claims processed last month. At the 0.01 significance level is it reasonable to conclude that the mean cost to process a claim is now less than $60? The sample information is reported below. **(4)** **CLO-2**

45, 49, 62, 40, 43, 61, 48, 53, 67, 63, 78, 64, 48, 54, 51, 56, 63, 69, 58, 51, 58, 59, 56, 57, 38, 76

(c) We have to accept or reject a large shipment of items. For quality control purposes, we collect a sample of 200 items and find 24 defective items in it. Construct a 95% confidence interval for the proportion of defective items in the whole shipment. **(4)** **CLO-2**

Q.5 **(a)** Anthony says to Eric that he is a stronger student because his average grade for the first six quizzes is higher. However, Eric replies that he is more stable because the variance of his grades is lower. The actual scores of the two friends (presumably, independent and normally distributed) are in the table. **(4)** **CLO-2**

Antony	85	92	97	65	75	96
Eric	81	79	76	84	83	77

State H_0 and H_1. Choose a suitable two-sample t-test and then conduct the test is there significant evidence to support Anthony's claim? State conclusions. Use the 5% level of significance

(b) The grades of a class of 9 students on a midterm report (x) and on the final examination (y) are as follows: **(8)** **CLO-2**

x	77	50	71	72	81	94	96	99	67
y	82	66	78	34	47	85	99	99	68

i. Construct scatter plot and comments.

ii. Estimate the linear regression line.

iii. Estimate the final examination grade of a student who received a grade of 85 on the midterm report.

iv. Compute correlation coefficient and comments.

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Statistical Annexure:

Cumulative Poisson Probability

X	$\mu = 0.2$
0	0.0187
1	0.9825
2	0.9989
3	0.9999

Cumulative Binomial Probability For n=10		
X	$p = 0.2$	$p = 0.8$
0	0.1074	-
1	0.3758	0.0000
2	0.6778	0.0001
3	0.8791	0.0009
4	0.9672	0.0064
5	0.9936	0.0328
6	0.9991	0.1209
7	0.9999	0.3222

Table: Critical values of Z

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.2	0.1151	0.1131	0.111	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-0.6	0.2451	0.2483	0.2514	0.2546	0.2578	0.2611	0.2643	0.2676	0.2709	0.2743
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	02810	0.2776
-0.4	0.3085	0.3050	03015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0	0.5	0.504	0.508	0.512	0.516	0.5199	0.5239	0.5279	0.5319	0.5359
0.2	0.5793	0.5832	0.5871	0.591	0.5948	0.5987	0.6026	0.6064	06103	06141
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952

Table: Critical values of the t-distribution

df	$t_{0.10}$	$t_{0.05}$	$t_{0.025}$	$t_{0.01}$	$t_{0.005}$
6	1.440	1.943	2.447	3.143	3.707
10	1.372	1.812	2.228	2.764	3.169
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779